

NEIGHBORHOOD COUNCIL APP



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(BSCS)**

Barani Institute of Information Technology

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A report submitted in partial fulfillment of the
requirements for the degree of

Bachelor of Science in Computer Science

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CERTIFICATE

It is certified that the contents and form of thesis entitled “**Neighborhood Council App**” submitted by **Muhammad Hasnat Nasim ,Siraj Hassan Khan, Muhammad Mushariq** have been found satisfactory for the requirement of the degree.

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ABSTRACT

The **Neighborhood Council App** is a mobile-based application designed to improve communication and collaboration between residents and neighborhood council authorities. Unlike traditional community management systems that rely on manual paperwork, physical visits, and delayed communication, this system provides a digital platform where residents can easily report issues, submit complaints, receive announcements, and stay informed about neighborhood activities and services.

The application provides a simple, interactive, and user-friendly interface suitable for people of all age groups. It enables users to communicate directly with council representatives, track the status of submitted complaints, and receive real-time notifications regarding community updates and events.

The system is developed using **Flutter** and **Android** technologies for mobile application development, **React.js** for the administration panel, **Node.js** for backend services, and **SQL Server** for database management. The application ensures secure data handling, efficient communication, and reliable information management. By digitizing neighborhood council operations, the system enhances transparency, promotes community participation, and improves the overall effectiveness of local governance.

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CHAPTER 1

INTRODUCTION

1.1 Introductory Background

The Neighborhood Council App was developed to provide residents with a convenient and efficient way to communicate with neighborhood council authorities and access community services. The system enables users to report local issues, submit complaints, receive announcements, and stay informed about neighborhood events and activities through a centralized digital platform. This application aims to improve transparency, community engagement, and service delivery by reducing the need for manual paperwork and physical visits to council offices. Residents can easily track the status of their requests, receive timely notifications, and participate in community-related matters. The app is designed to support people of all age groups by offering a simple, interactive, and user-friendly interface for effective communication between citizens and local authorities.

1.2 Problem Statement

Traditional neighborhood council management relies on manual processes, physical paperwork, and in-person visits, resulting in inefficient communication and service delivery. The key problems include the lack of a centralized digital platform for reporting issues and receiving announcements, the absence of real-time complaint tracking, and the use of outdated manual election systems. Additionally, there is no digital tool for committee management, no transparency in budget allocation, no streamlined process for changing council membership, and no centralized notification system for real-time updates. These challenges collectively hinder resident participation, reduce accountability, and limit the overall effectiveness of neighborhood governance.

1.3 Proposed Solution

This application can be used by anyone who needs help managing tasks efficiently. It is particularly useful for elderly people, patients with memory issues, or individuals with busy schedules. The system can be expanded in the future to include cloud

synchronization, AI-based voice recognition, and cross-platform integration.

Universities for managing academic advising across multiple faculties.

Colleges to streamline advisor-student interactions and improve academic support.

Vocational and Technical Institutes to provide tailored course guidance.

1.4 Objectives

The main objectives of the Neighborhood Council App are as follows:

Council Management

To allow the Admin to create and manage neighborhood councils using Google Maps integration.

To enable the Admin to announce elections and manage council change requests submitted by members.

Election System

To allow Members to vote for candidates (President, Vice President, Treasurer) through a secure digital Election system.

To enable Members to form and register election panels.

To automatically calculate and display election results based on total votes received.

Complaint Management

To allow Members to submit complaints with titles, descriptions, and optional images.

To enable Presidents and Vice Presidents to view pending complaints and form committees to resolve them.

To allow committee members to upload meeting minutes (PDF) and resolution images as proof of work.

To enable Presidents to review committee reports and approve or reject complaints.

Budget and Treasury Management

To allow Presidents to approve budget requests for complaint resolution.

To enable Treasurers to release approved budgets from the Council Treasury Funds.

To maintain a transparent record of all fund releases through the Council Treasury Fund Releases table.

Member Management

To allow Members to join multiple councils and change their council membership.

To enable Members to view the status of their complaints in real-time (Pending, In Progress, Committee Resolved, Approved, Rejected, Resolved).

To allow Members to submit suggestions to the council.

Notification System

To provide real-time notifications to users regarding complaint status updates, election announcements, and council changes.

To allow Admins and Presidents to send announcements to all council members.

Security and Data Protection

To ensure secure user authentication using JWT (JSON Web Tokens) .

To protect sensitive user data (CNIC, phone numbers, addresses) through encrypted storage and role-based access control.

User-Friendly Interface

To provide a simple, interactive, and user-friendly interface suitable for users of all age groups.

To ensure the application is accessible on mobile devices (Android and iOS) through Flutter.

CHAPTER 2

PROJECT BACKGROUND

2.1 Related Project and Research Articles

This section contains discussion about the related projects like this project which was designed in the past by any other student of BIIT and the discussion about the designing of that project.

A **Complaint Management System** was developed in 2023 which allowed users to submit complaints and track their status as pending or resolved. The system provided a basic platform for issue reporting and allowed users to view the progress of their submitted complaints. However, the project did not support multiple user roles such as Admin, President, or Treasurer, which limited its ability to implement structured governance workflows. Additionally, it lacked election systems, committee formation capabilities, and budget management features, making it insufficient for managing complete neighborhood council operations.

The screenshot displays the 'COMPLAINT MANAGEMENT SYSTEM' interface. At the top, there is a yellow header bar with the system name on the left and a 'Logout' button on the right. A dark blue sidebar on the left contains a user profile for 'John Smith' with a deer icon, and a list of navigation options: Dashboard, Account Setting, Lodge Complaint, and Complaint History. The main content area is titled '> Profile info' and shows 'John Smith's Profile'. It includes a 'Last Updated at' timestamp of '2020-05-08 22:16:22'. The form contains several input fields: 'Full Name' (John Smith), 'User Email' (john@gmail.com), 'Contact' (9999999999), 'Address' (empty), 'State' (dropdown menu), 'Country' (empty), 'Pincode' (empty), and 'Reg Date' (2020-05-08 22:10:50). A blue 'Submit' button is located at the bottom of the form. The footer of the page is light blue and contains the copyright notice '© 2020 CMS All rights reserved.' and a small upward arrow icon.

Figure 2.1: Complaint Management System

Community Notification App was developed in 2024 which allowed admins to send announcements and push notifications to community members. This application focused on one-way communication and provided a simple interface for broadcasting important updates to residents. However, the project lacked complaint management features, which meant residents could not report issues or track their resolution. It also did not include role-based governance or election systems, which restricted its functionality to basic information dissemination rather than interactive community management.

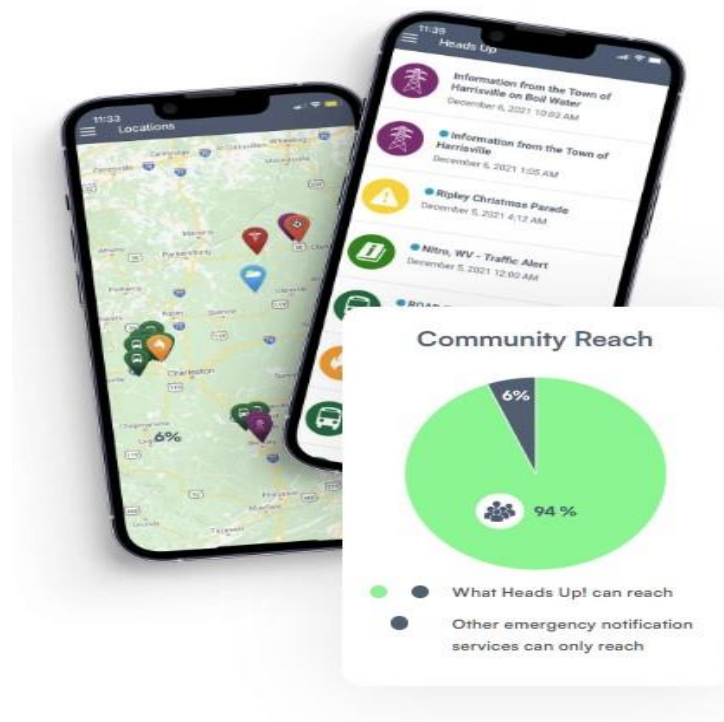
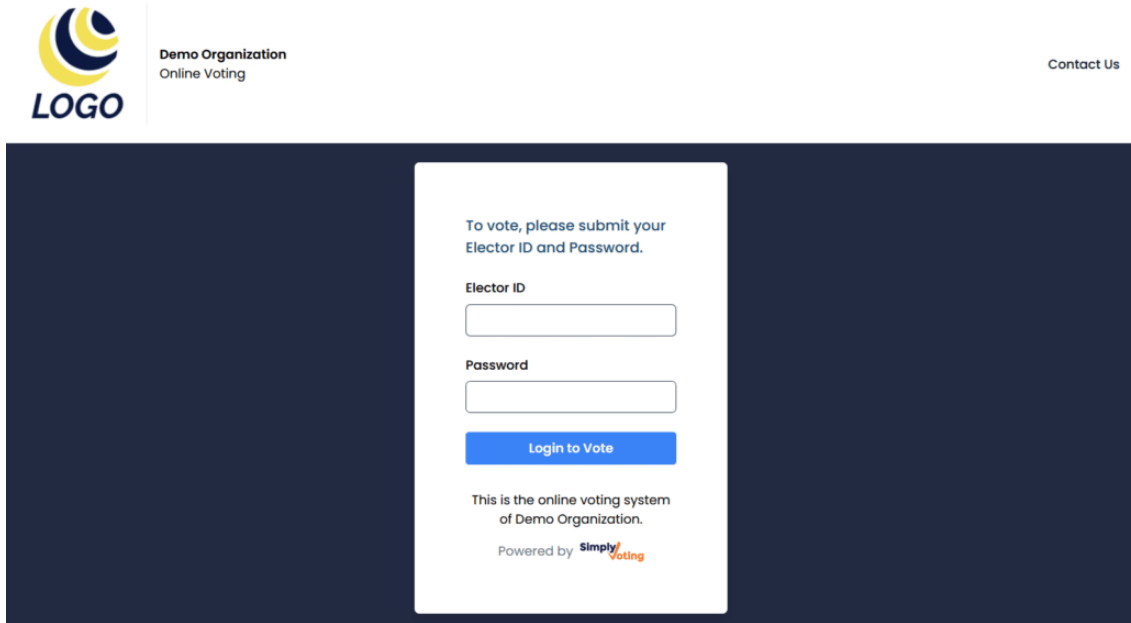


Figure 2.2: Community Notification App

An **Online Voting System** was developed in 2024 which supported candidate registration and vote counting. The system provided a digital platform for conducting elections and automatically calculating results. However, this project did not include council management features such as creating councils or managing memberships. It also lacked complaint handling and committee formation features, which meant it could only be used for electoral processes and could not support the broader governance activities required by neighborhood councils.



The screenshot displays the login interface of the Online Voting System. At the top left, there is a logo consisting of a yellow and blue circular emblem above the word "LOGO" in blue. To the right of the logo, the text "Demo Organization" and "Online Voting" is displayed. In the top right corner, a "Contact Us" link is visible. The main content area has a dark blue background. A white rectangular box in the center contains the login instructions: "To vote, please submit your Elector ID and Password." Below this, there are two input fields: "Elector ID" and "Password". A blue button labeled "Login to Vote" is positioned below the password field. At the bottom of the white box, it states "This is the online voting system of Demo Organization." and "Powered by Simply Voting", with "Simply Voting" featuring a red and orange logo.

Figure 2.3: Online Voting System

Fix My Street is a web-based application that allows citizens to report local issues to municipal authorities. Its key features include issue reporting where users can submit complaints about potholes, garbage, streetlights, and other local problems; location tracking where complaints are pinned to a map for easy visualization; status updates allowing users to track the progress of their complaints through stages such as Pending, In Progress, and Resolved; and image upload which enables users to attach photos to their complaints for better documentation. However, Fix My Street does not support election systems, budget management, committee formation, or role-based governance. It is designed for municipal-level reporting rather than neighborhood council operations, which limits its applicability to the comprehensive governance needs of local councils.

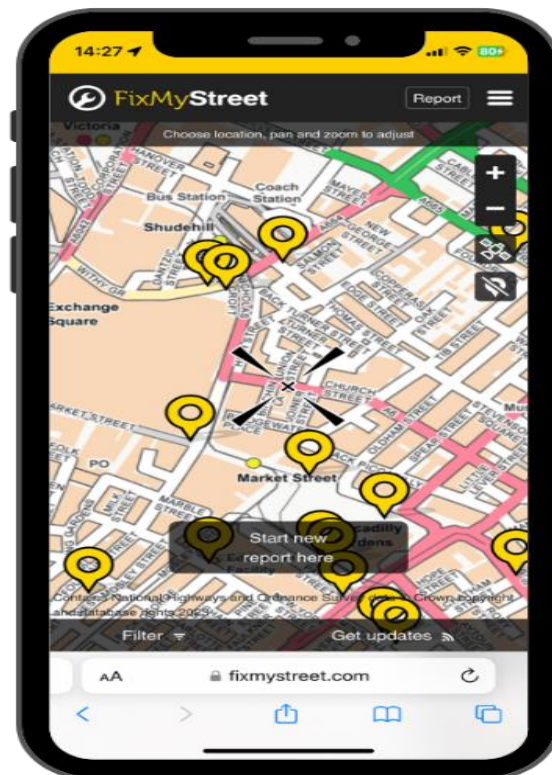


Figure 2.4: Fix My Street

CHAPTER 3

CONCEPTUAL DESIGN

This chapter describes how the requirements were gathered and specifies the functional and non-functional requirements of the system. It also includes the conceptual diagram, data flow diagram, and entity relationship diagram of the application.

3.1 Requirements Elicitation

The question is that from where the requirements are gathered? The requirements for the Neighborhood Council App were gathered from the Director of BIIT and from the project supervisor, Mr. Shahid Abid. In each meeting, there was discussion of assigning the objectives and requirements for the system. The scope and specific project requirements were defined based on the needs of neighborhood council governance, including complaint management, election systems, committee formation, and budget tracking. Throughout the semester, the project team adhered to the instructions of the committee and supervisor, systematically addressing the tasks outlined in the project plan. The development process was organized into sub-modules, with progress reviewed and adjusted during regular meetings. This structured approach ensured that the application met all specified requirements and effectively addressed the needs of residents, council members, and administrators.

3.2 Requirements Specification

A requirement specification is a collection of the set of all requirements that are to be imposed on the design and verification of the product. The specification also contains other related information necessary for the design, verification, and maintenance of our application.

3.2.1 Functional Requirements

Admin:

- Create and manage neighborhood councils using Google Maps integration

- Announce elections and manage election schedules
- Accept or reject council change requests from members

Member:

- Register and login to the application
- Submit complaints with title, description, and optional images
- Track the status of submitted complaints (Pending, In Progress, Committee Resolved, Approved, Rejected, Resolved)
- Vote in elections for President, Vice President, and Treasurer
- Join multiple neighborhood councils
- Request to change council membership
- Join committees formed by the President
- Submit suggestions to the council

President:

- View all complaints in the council
- Form committees to resolve complaints
- Assign members to committees
- Review committee reports including meeting minutes and resolution images
- Approve or reject complaints
- Approve budget requests for complaint resolution

Vice President:

- Same role and privileges as the President

Treasurer:

- View approved complaints

- Release budgets for approved complaints
- Manage council treasury funds
- View total budget, released amount, and available amount.

3.2.2 Non-Functional Requirements

- Scalability: The ability to scale the application to accommodate an increasing number of users and councils, as well as an increasing volume of complaints and transactions.
- Security: The ability to secure data and prevent unauthorized access through JWT-based authentication and role-based access control.
- Privacy: The ability to protect the privacy of users and ensure that data is collected, processed, and stored in accordance with privacy regulations.
- Reliability: The ability to ensure that the application is highly available and operates with a high degree of uptime, even in the face of network outages or other technical difficulties.
- Compatibility: The ability to run the application on a wide range of mobile devices and ensure that it works seamlessly with other software components.
- Usability: The application must have an intuitive and user-friendly interface, ensuring that users of all age groups can easily navigate and utilize its features with minimal training or support.
- Performance: The application must provide fast and responsive interactions, with minimal latency in retrieving and updating data.
- Maintainability: The application should be designed for easy maintenance and updates, with well-documented code and modular architecture.

3.2.3 Domain Requirements

The domain requirements describe constraints and conditions specific to the neighborhood council governance domain.

- **Performance:** The application must provide fast and responsive interactions, with minimal latency in retrieving and updating data. It should handle a large number of concurrent users without performance degradation.
- **Scalability:** The application should be designed to accommodate growth, both in terms of user volume and data size, ensuring it can handle increased load as more councils and members use the system.
- **Availability:** The application must be highly available, with minimal downtime. It should include backup and recovery mechanisms to ensure data integrity and availability in case of system failures.
- **Usability:** The application must have an intuitive and user-friendly interface, ensuring that both residents and council members can easily navigate and utilize its features with minimal training or support.
- **Security:** The application must implement robust security measures to protect sensitive data. This includes encryption for data transmission and storage, secure user authentication, and authorization mechanisms to restrict access based on user roles.
- **Maintainability:** The application should be designed for easy maintenance and updates, with well-documented code and modular architecture that allows for efficient bug fixes, enhancements, and feature additions.
- **Legal Compliance:** The application shall comply with all applicable laws and regulations regarding data protection, privacy, and local governance standards.
- **Local Governance Standards:** The application shall support the standard processes and workflows of neighborhood council governance as defined by local authorities.
- **Language Support:** The application interface shall be available in English

and optionally in local languages to ensure accessibility for all residents.

- **Offline Support:** The application shall provide limited offline functionality, allowing users to view previously loaded data when internet connectivity is unavailable.

3.3 Conceptual Diagram

The conceptual diagram of the Neighborhood Council App provides a high-level visual representation of the system, illustrating the fundamental components and their interactions. It serves as a blueprint that shows how different users interact with the system, how data moves between components, and how core functionalities such as council creation, complaint submission, election voting, committee formation, report submission, complaint review, and budget release are connected to one another. The diagram highlights the complete workflow of the system, starting from the Admin creating councils using Google Maps, Members submitting complaints and voting in elections, Presidents forming committees and reviewing reports, and Treasurers releasing budgets for approved complaints. Real-time notifications are sent to users at each stage to ensure transparency and efficient communication throughout the governance process.



Figure 3.1: conceptual diagram of the Neighborhood Council App

Approved By: Mr. Shahid Abid

3.4 Data Flow Diagram

The Data Flow Diagram (DFD) of the Neighborhood Council App illustrates the movement of data between external entities, processes, and data stores within the system. It begins with the Login process where Admin, President, Vice President, Treasurer, and Member provide their credentials, which are verified against the Users data store. The Admin creates councils by sending council data to the NHC Zones data store. Members interact with the system by submitting complaints, which are stored in the Complaints data store, and voting in elections, which updates the Elections data store. Members also track the status of their complaints by retrieving information from the Complaints data store. The President forms committees by accessing the Committees data store, and committee members submit reports that update the Complaints data store. The Treasurer releases budgets by updating the Treasury data store. Finally, the system sends notifications to users through the Notifications data store, ensuring real-time communication at every stage of the governance process.

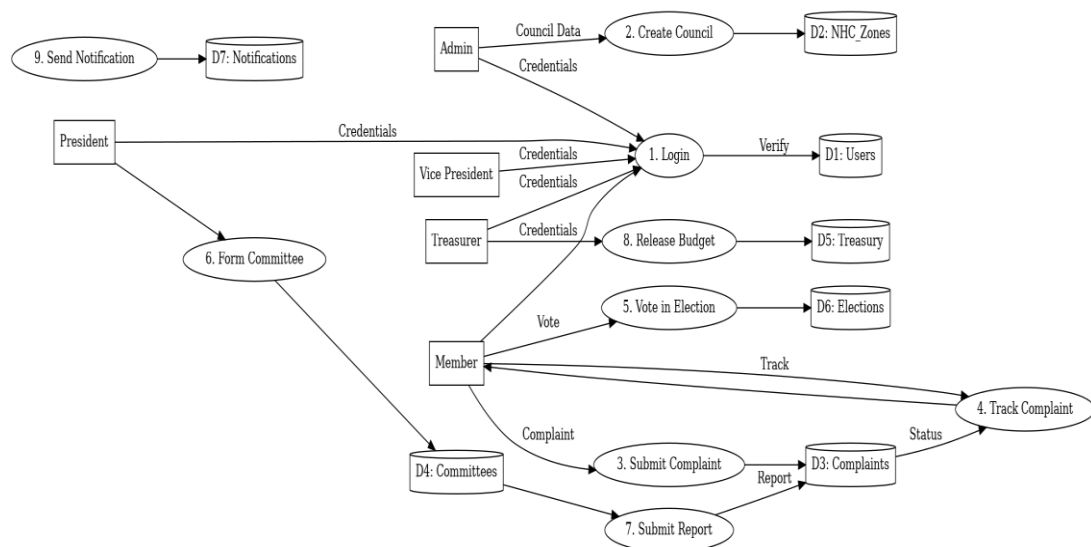


Figure 3.2: Data Flow Diagram

Approved By: Mr. Shahid Abid

3.5 Database Design ERD

The Entity Relationship Diagram (ERD) of the Neighborhood Council App illustrates the database structure, showing entities, attributes, and their relationships. The **USERS** entity acts as the central hub, linking to **USERNHCS** for multi-council membership, **COMPLAINTS** for issue reporting, and **ELECTIONS** for voting. The **NHC_ZONES** entity represents neighborhood councils and connects to complaints, committees, and treasury funds. The complaint resolution workflow is managed through relationships between **COMPLAINTS**, **COMMITTEES**, and **COUNCIL_TREASURY_FUND_RELEASES**. The election process involves **ELECTIONS**, **CANDIDATES**, **NOMINATIONS**, and **PANELS**, ensuring structured governance. This ERD provides a complete data model for the system, ensuring data integrity and efficient management of neighborhood council operations.

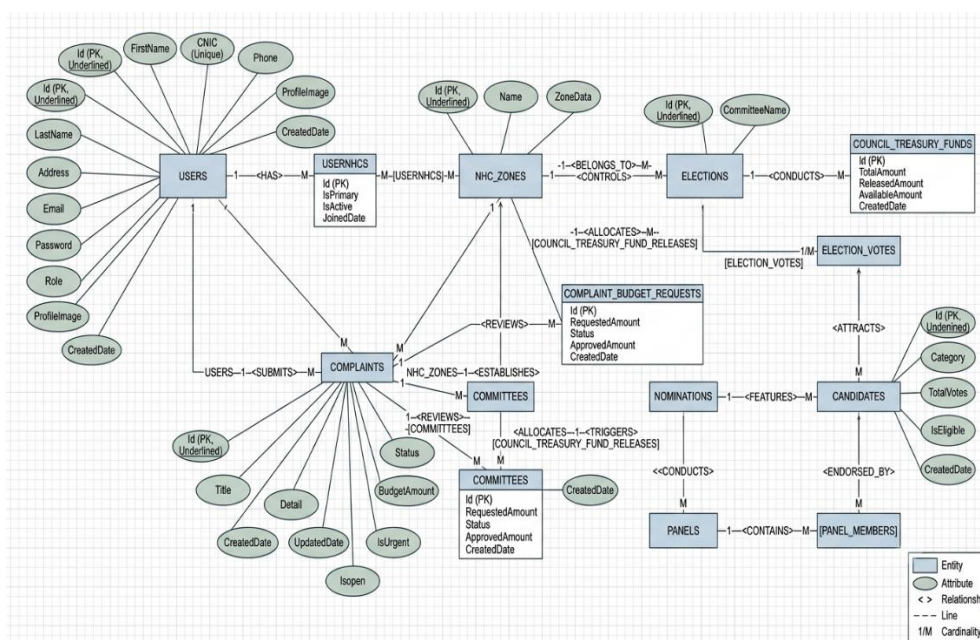


Figure 3.3: Entity Relationship Diagram (ERD)

Approved By: Mr. Shahid Abid

CHAPTER 4

IMPLEMENTATION

This chapter includes the discussion about the Tools and Technologies which have been used for designing of this application. It also includes how the application has been designed through coding. Also, this chapter includes the discussion about graphical user interface, and this can be shown through project screen shots.

4.1 Tools & Technologies

Following Tools and Technologies are used to develop this application.

Flutter

Flutter is an open-source UI software development kit created by Google. It is used to develop cross-platform applications for Android, iOS, and web from a single codebase. Flutter combines ease of development with performance similar to native performance while maintaining visual consistency between platforms. The framework uses the Dart programming language and includes a rich set of pre-built widgets that make UI development fast and efficient. For the Neighborhood Council App, Flutter is used to build the mobile application that runs on both Android and iOS devices.

React.js

React.js is a JavaScript library for building user interfaces developed by Facebook. It is used for the admin panel of the Neighborhood Council App. React allows developers to create reusable UI components and manage application state efficiently. The admin panel provides council administrators with tools to manage councils, elections, and user requests. React's component-based architecture makes it easy to maintain and extend the admin panel as new features are added.

Microsoft SQL Server

Microsoft SQL Server is a relational database management system developed by Microsoft. As a database server, it is a software product with the primary function of storing and retrieving data as requested by other software applications which may run either on the same computer or on another computer across a network. SQL Server is used as the database for the Neighborhood Council App, storing all application data including user information, councils, complaints, committees, treasury funds, and

election records. The database schema includes 23 tables that define the complete data model for the application.

Node.js

Node.js is an open-source, cross-platform JavaScript runtime environment that executes JavaScript code outside of a web browser. It is used as the backend server for the Neighborhood Council App. Node.js provides a fast and scalable server-side environment that handles API requests, authentication, and data processing. The backend is built using the Express.js framework, which provides a robust set of features for building RESTful APIs.

Visual Studio Code

Visual Studio Code is a lightweight but powerful source code editor developed by Microsoft. It is used as the integrated development environment (IDE) for developing the Neighborhood Council App. Visual Studio Code provides features such as syntax highlighting, code completion, debugging, and Git integration. It supports multiple programming languages and frameworks including Flutter, React.js, and Node.js.

4.2 Pseudo Code

Pseudo code is a simplified, informal way of describing an algorithm or program logic. It uses plain language mixed with programming constructs to explain how a system works without requiring strict syntax. Below is the pseudo code for key processes in the Neighborhood Council App.

User Login Process:

- User enters CNIC and Password
- System validates credentials against Users table
- IF (CNIC and Password match a record in Users table) THEN
- Generate JWT token
- Retrieve user role (Admin, Member, President, Vice President, Treasurer)

- Redirect user to role-specific dashboard
- ELSE
- Display error message: "Invalid CNIC or Password"
- END IF

Admin – Create Council Process:

- Admin logs in and accesses Admin Dashboard
- Admin clicks "Create Council" button
- Google Maps interface opens
- Admin selects a location on the map
- Admin enters council name
- Admin clicks "Save"
- System inserts new record into NHC_Zones table
- System displays success message: "Council created successfully"
- Council is now available for members to join

Member – Submit Complaint Process:

- Member logs in and accesses Member Dashboard
- Member clicks "Submit Complaint" button
- Member enters complaint title and description
- Member optionally uploads images
- Member clicks "Submit"
- System inserts new record into Complaints table
- Complaint status is set to "Pending"
- System inserts complaint images into Complaint Images table (if uploaded)

- System sends notification to President of the council
- System displays success message: "Complaint submitted successfully"

Member – Track Complaint Status Process:

- Member logs in and accesses Member Dashboard
- Member clicks "My Complaints" button
- System retrieves all complaints submitted by the member from Complaints table
- System displays list of complaints with their current status
- Status options: Pending, In Progress, Committee Resolved, Approved, Rejected, Resolved
- Member can click on any complaint to view details
- Member can see committee remarks and president remarks if available

Member – Vote in Election Process:

- Member logs in and accesses Member Dashboard
- Member clicks "Elections" button
- System checks if an active election exists for the member's council
- IF (active election exists) THEN
- System displays list of candidates for President, Vice President, and Treasurer
- Member selects a candidate for each position
- Member clicks "Submit Vote"
- System inserts record into Election Votes table
- System updates Total Votes for each selected candidate in Candidates table

- System displays success message: "Vote submitted successfully"
- ELSE
- Display message: "No active election at this time"
- END IF

President – View Complaints Process:

- President logs in and accesses President Dashboard
- President clicks "Complaints" button
- System retrieves all complaints for the president's council from Complaints table
- System displays complaints grouped by status
- President can filter complaints by status: Pending, In Progress, Committee Resolved, Approved, Rejected, Resolved
- President can click on any complaint to view full details including images

President – Form Committee Process:

- President logs in and accesses President Dashboard
- President clicks "Pending Complaints" button
- President selects a complaint from the list
- President clicks "Form Committee" button
- System opens "Create Committee" form
- President enters committee name
- President selects members from the council to join the committee
- President designates one member as committee head (Is Head = 1)
- President clicks "Create Committee"
- System inserts new record into Committees table
- System inserts records into Committee Members table for each selected

member

- System updates Complaint status to "In Progress"
- System updates Complaint table with Committee ID
- System sends notifications to all committee members
- System displays success message: "Committee formed successfully"

Committee – Submit Report Process:

- Committee member logs in and accesses Committee Dashboard
- Committee member clicks "My Committees" button
- System displays list of committees the member is assigned to
- Committee member selects a committee
- Committee member clicks "Submit Report" button
- Committee member uploads meeting minutes as PDF file
- Committee member uploads resolution images
- Committee member enters remarks (optional)
- Committee member clicks "Submit"
- System inserts record into Complaint Meetings table with PDF file path
- System inserts records into Complaint Resolution Images table with image paths
- System updates Complaint status to "Committee Resolved"
- System sends notification to President for review
- System displays success message: "Report submitted successfully"

President – Approve or Reject Complaint Process:

- President logs in and accesses President Dashboard
- President clicks "Committee Resolved Complaints" button

- System displays complaints that are in "Committee Resolved" status
- President selects a complaint
- President reviews committee report, meeting minutes, and resolution images
- President enters remarks
- President clicks one of the following:
 - "Approve" – Complaint status updated to "Approved"
 - "Reject" – Complaint status updated to "Rejected"
- IF (Approved) THEN
 - System creates budget request in Complaint Budget Requests table
 - System sends notification to Treasurer for budget release
- ELSE IF (Rejected) THEN
 - System sends notification to member with rejection reason
- END IF
- System displays success message: "Complaint reviewed successfully"

Treasurer – Release Budget Process:

- Treasurer logs in and accesses Treasurer Dashboard
- Treasurer clicks "Pending Budget Requests" button
- System retrieves all pending budget requests from Complaint Budget Requests table
- Treasurer selects a budget request
- Treasurer views complaint details and approved amount
- Treasurer enters released amount (may be less than requested)
- Treasurer enters remarks (optional)
- Treasurer clicks "Release Budget"

- System updates ComplaintBudgetRequests status to "Released"
- System updates ApprovedAmount with released amount
- System inserts record into CouncilTreasuryFundReleases table
- System updates CouncilTreasuryFunds:
- Released Amount = ReleasedAmount + released amount
- AvailableAmount = AvailableAmount - released amount
- System updates Complaint status to "Resolved"
- System updates Complaint with BudgetAmount and ResolvedByCNIC
- System sends notification to member: "Your complaint has been resolved"
- System displays success message: "Budget released successfully"

Member – Request Council Change Process:

- Member logs in and accesses Member Dashboard
- Member clicks "Change Council" button
- System displays current council and list of available councils
- Member selects a new council from the list
- Member enters a reason for the change (optional)
- Member clicks "Submit Request"
- System inserts new record into Requests table
- Request status is set to "Pending"
- System sends notification to Admin for approval
- System displays success message: "Request submitted successfully"

Admin – Accept or Reject Council Change Request Process:

- Admin logs in and accesses Admin Dashboard

- Admin clicks "Change Requests" button
- System retrieves all pending requests from Requests table
- Admin selects a request
- Admin reviews the request details
- Admin enters remarks (optional)
- Admin clicks one of the following:
- "Accept" – Request status updated to "Accepted"
- "Reject" – Request status updated to "Rejected"
- IF (Accepted) THEN
- System updates UserNHCs table:
- IsActive = 0 for current council
- Insert new record for new council with IsPrimary = 1
- System sends notification to member: "Council change approved"
- ELSE IF (Rejected) THEN
- System sends notification to member: "Council change request rejected"
- END IF
- System displays success message: "Request processed successfully"

4.3 Graphical User Interface

Following are the screenshots of the Neighborhood Council App. Each screen is designed to provide an intuitive and user-friendly experience for all user roles.

Member Login Screen

The login screen allows users to access the application using their CNIC and password. Users are redirected to their role-specific dashboard upon successful authentication.

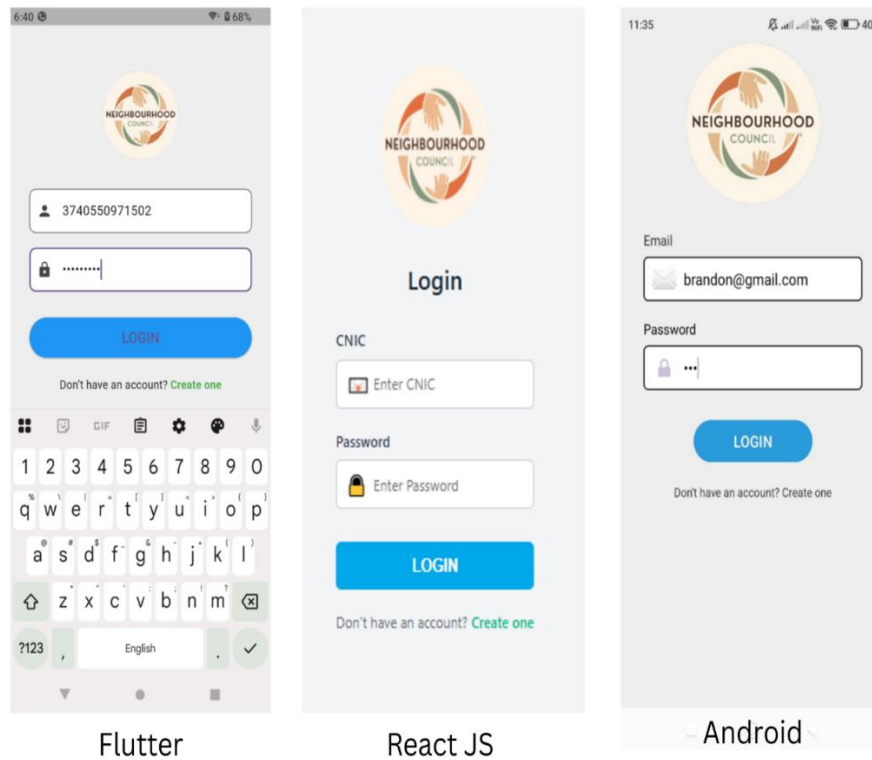


Figure 4.1: Login Screen

Member Dashboard

The Member Dashboard displays all key features available to a member including submitting complaints, tracking complaint status, voting in elections, joining committees, and changing council membership.

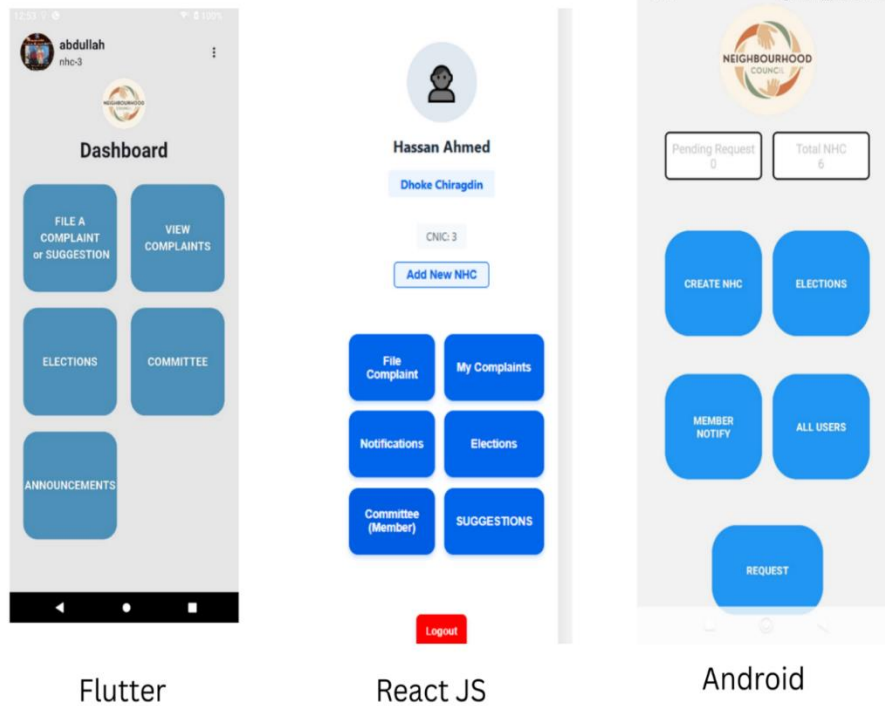
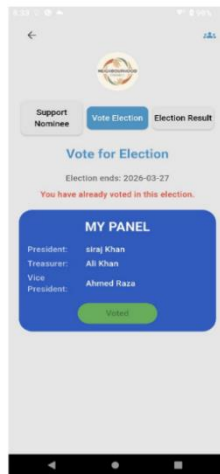


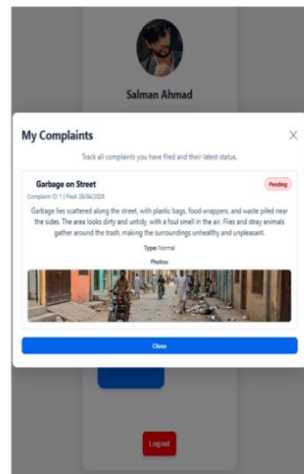
Figure 4.2: Member Dashboard Screen

Election Voting Screen

Members can vote in council elections by selecting candidates for President, Vice President, and Treasurer. The system ensures each member votes only once per election.



Flutter



React JS



Android

Figure 4.3: Election Voting Screen

Admin Dashboard

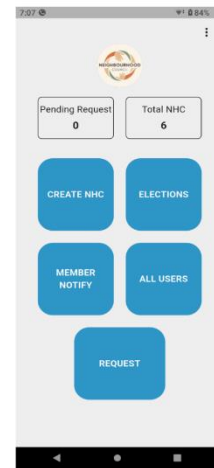
The Admin Dashboard displays all administrative features including council management, election announcements, and council change requests.



Flutter



React JS



Android

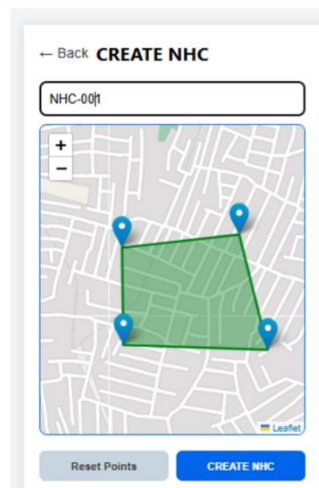
Figure 4.4: Admin Dashboard

Create Council Screen

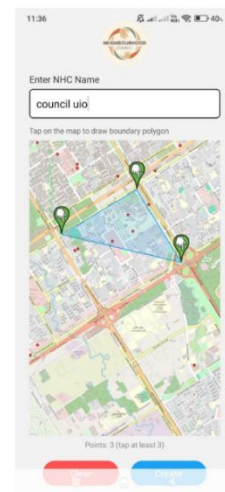
The Admin can create new neighborhood councils using Google Maps integration. The Admin selects a location on the map and enters a council name



Flutter



React JS



Android

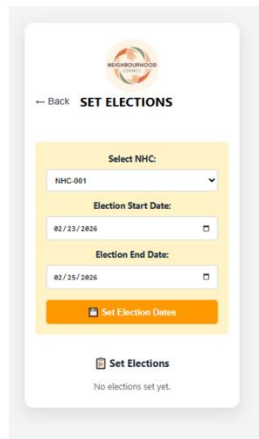
Figure 4.5: Create NHC

Admin – Election Announcement

The Admin can announce elections for each council by setting the election start date, end date, and nomination period.



Flutter



React JS

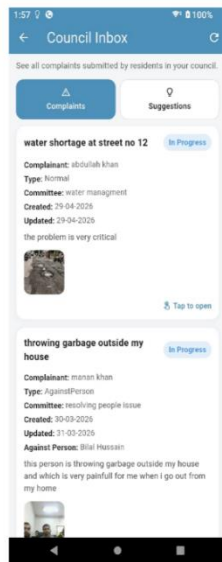


Android

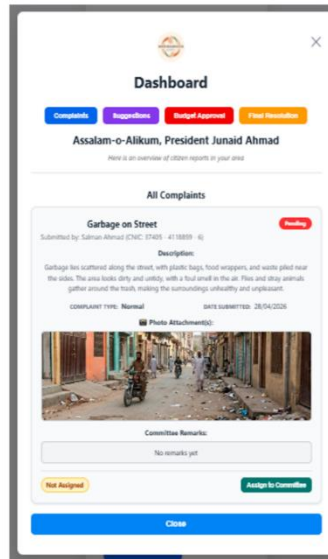
Figure 4.6: Election Announcement

President – Complaint Management

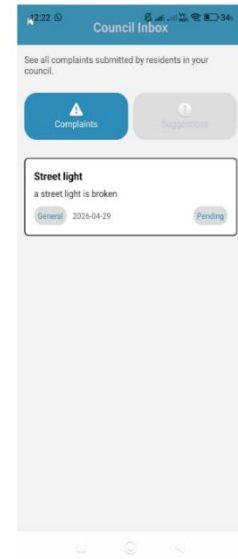
The President can view all complaints within their council, filter by status, and select complaints to form committees.



Flutter



React JS

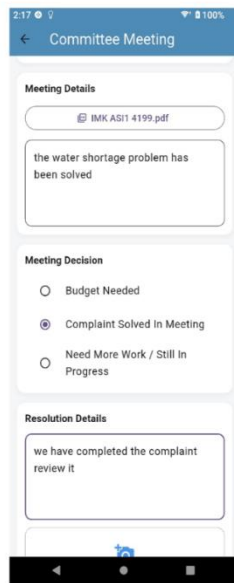


Android

Figure 4.7: Complaint Management

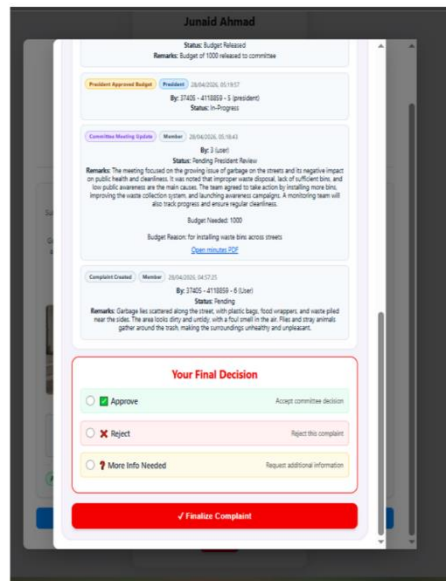
President – Form Committee

The President can form committees by selecting a complaint, entering a committee name, and assigning members from the council.



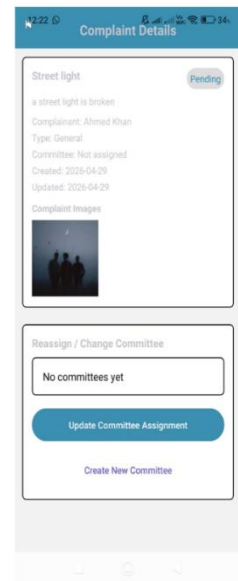
The Flutter interface shows a 'Committee Meeting' screen. It includes a 'Meeting Details' section with a PDF attachment 'IMK ASI1 4199.pdf' and a text box stating 'the water shortage problem has been solved'. Below this is a 'Meeting Decision' section with three radio buttons: 'Budget Needed', 'Complaint Solved In Meeting' (which is selected), and 'Need More Work / Still In Progress'. At the bottom is a 'Resolution Details' section with a text box containing 'we have completed the complaint review it'.

Flutter



The React JS interface displays a detailed view of a committee meeting for 'Junaid Ahmad'. It shows the status 'Budget Released', a budget of 1000, and a list of members including the President and a committee member. The 'Remarks' section describes a meeting about garbage on the streets. At the bottom, there is a 'Your Final Decision' section with three options: 'Approve' (selected), 'Reject', and 'More Info Needed'. A red 'Finalize Complaint' button is at the very bottom.

React JS



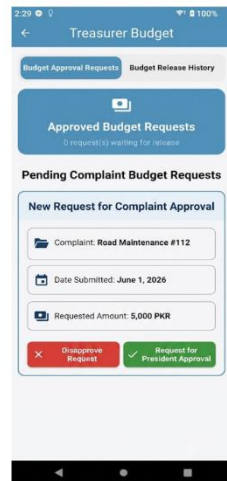
The Android interface shows a 'Complaint Details' screen for a 'Street light' complaint. It indicates the status is 'Pending' and lists details such as the complainant 'Ahmed Khan', the type 'General', and the committee status 'Not assigned'. It also shows the creation and update dates as '2025-04-29'. Below the details is a 'Complaint Images' section with a photo of a street at night. At the bottom, there is a 'Reassign / Change Committee' section with a 'No committees yet' message, an 'Update Committee Assignment' button, and a 'Create New Committee' link.

Android

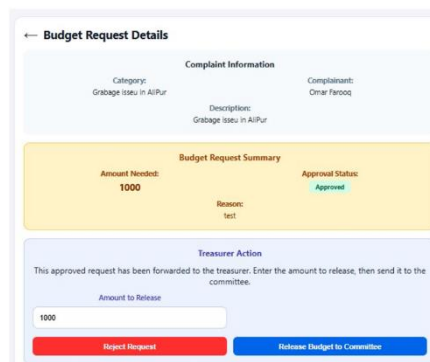
Figure 4.8: Form Committee

Treasurer – Budget Release Screen

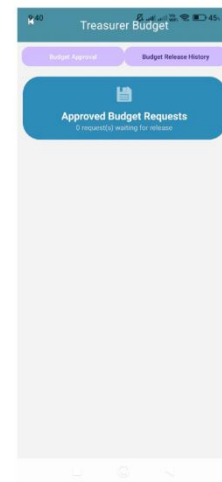
The Treasurer can view pending budget requests and release funds for approved complaints. The released amount is deducted from the council treasury.



Flutter



React JS

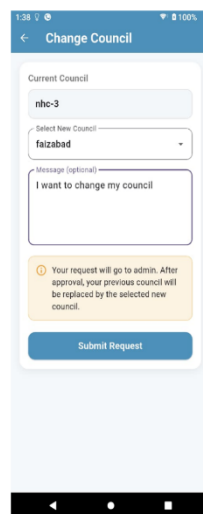


Android

Figure 4.9: Budget Release Screen

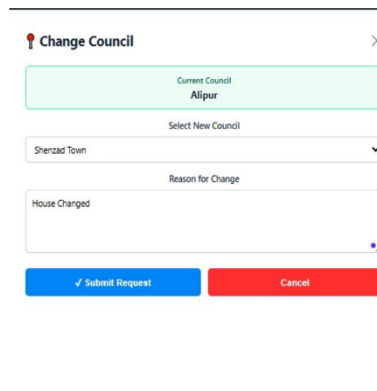
Member – Change Council Request

Members can request to change their council membership by selecting a new council from the available list. The request is sent to the Admin for approval.



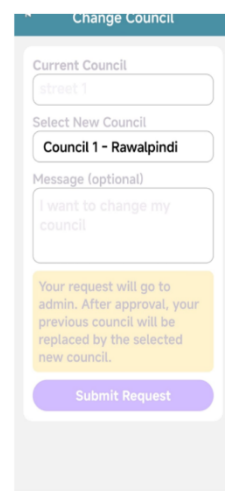
The Flutter screen shows a 'Change Council' form. It includes a 'Current Council' field with the value 'nho-3', a 'Select New Council' dropdown menu with 'faizabad' selected, and a 'Message (optional)' text area containing 'I want to change my council'. A yellow informational box states: 'Your request will go to admin. After approval, your previous council will be replaced by the selected new council.' At the bottom is a blue 'Submit Request' button.

Flutter



The React JS screen displays a 'Change Council' modal. It features a 'Current Council' field with 'Alipur', a 'Select New Council' dropdown with 'Shenzad Town' selected, and a 'Reason for Change' text area with 'House Changed'. A yellow informational box contains the text: 'Your request will go to admin. After approval, your previous council will be replaced by the selected new council.' The bottom has two buttons: a blue 'Submit Request' button and a red 'Cancel' button.

React JS



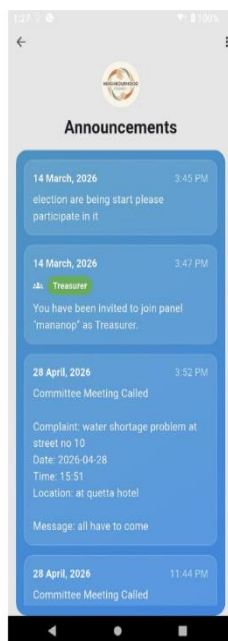
The Android screen shows a 'Change Council' form. It has a 'Current Council' field with 'street 1', a 'Select New Council' dropdown with 'Council 1 - Rawalpindi' selected, and a 'Message (optional)' text area with 'I want to change my council'. A yellow informational box states: 'Your request will go to admin. After approval, your previous council will be replaced by the selected new council.' At the bottom is a purple 'Submit Request' button.

Android

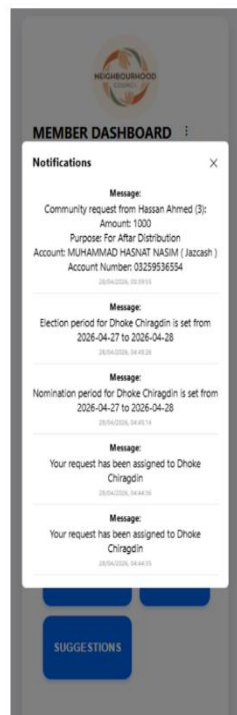
Figure 4.10: Change Council Screen

Member – View Notifications

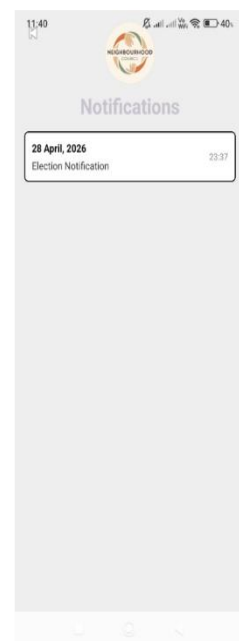
Members receive real-time notifications regarding complaint status updates, election announcements, and council news.



Flutter



React JS



Android

Figure 4.11: Notification Screen

CHAPTER 5

CONCLUSION

The concluding remarks and future directions for the development and up gradation of the existing projects are given below. Some of the limitations of the project are also given.

5.1 Conclusion Remarks

The Neighborhood Council App successfully provides a comprehensive digital platform for local governance. The system enables residents to report issues, participate in elections, join committees, and manage council memberships. The Admin can create councils using Google Maps and manage elections. Presidents and Vice Presidents can effectively manage complaints by forming committees and reviewing reports. Treasurers can release budgets for approved complaints and manage treasury funds. The application enhances transparency, accountability, and community participation in neighborhood governance. The database design supports multiple councils, multi-council membership, and a complete election system with panels, nominations, and voting. The system is developed using Flutter for the mobile application, React.js for the admin panel, Node.js for backend services, and SQL Server for database management.

The Neighborhood Council App fills the gap identified in existing applications by integrating complaint management, role-based governance, digital elections, committee formation, budget tracking, and multi-council support into one comprehensive platform. The application has been tested and found to be stable, responsive, and user-friendly.

5.2 Future Directions

The Neighborhood Council App has the potential to evolve further, offering even more sophisticated features to enhance neighborhood governance. Future development could focus on the following areas:

- **AI-based Complaint Categorization:** Artificial intelligence can be used to

automatically classify complaints by type and priority, helping Presidents prioritize urgent issues and allocate resources efficiently.

- **Real-time GPS Tracking:** Integration of real-time GPS tracking can allow members to see the location of complaints on a map and track the progress of committee members working on resolution.
- **Multilingual Support:** Implementing multilingual support can make the application more accessible to diverse communities, ensuring all residents can participate in neighborhood governance regardless of their language.
- **Integration with Government Portals:** Connecting the application with city or municipal complaint systems can provide a more holistic approach to community problem-solving and enable seamless data sharing between different levels of government.
- **Chat System:** A direct messaging system between members and council representatives can improve communication and enable faster resolution of minor issues without requiring formal complaint submission.
- **Payment Gateway Integration:** Online payment gateway integration can enable residents to contribute to council funds or pay for services directly through the application.
- **Mobile App for Committee Members:** A dedicated mobile interface for committee members can make it easier for them to upload reports, images, and meeting minutes while on the go.
- **Analytics Dashboard:** An analytics dashboard for Admins and Presidents can provide insights into complaint trends, election participation rates, and budget utilization, enabling data-driven decision-making.

5.3 Limitations

Following are the limitations of the Neighborhood Council App.

- **Internet Dependency:** The application requires an active internet connection for real-time updates, complaint submission, and notification

delivery. Users in areas with poor connectivity may experience delays or limited functionality.

- **User Adoption:** The success of the application depends on the level of user adoption. Factors such as digital literacy, ease of use, and perceived benefits may affect how quickly residents and council members embrace the system.
- **Data Privacy and Security:** The application stores sensitive user information including CNIC, phone numbers, and addresses. While security measures such as JWT authentication and data encryption have been implemented, continuous monitoring and updates are required to protect against emerging threats.
- **Scalability:** The application is designed to handle multiple councils and thousands of users. However, as the number of councils and users grows, server upgrades and database optimization may be required to maintain performance.
- **Google Maps API Cost:** The Google Maps API is used for council creation. High usage may incur costs, which could become a consideration for large-scale deployment.
- **Manual Committee Formation:** Committee members must be selected manually by the President. Future versions could include automated committee assignment based on member expertise or availability.
- **No Offline Support:** The application does not support full offline functionality. Users cannot submit complaints or view updates when offline. Limited offline caching could be implemented in future versions.

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ANNEXURE

This section contains supplementary information including step-by-step processes for authentication, user dashboards, and a summary of the database schema.

Annexure A: Authentication Process

The authentication process allows users to securely access the Neighborhood Council App based on their assigned role.

- Step 1: User opens the application and navigates to the Login Screen.
- Step 2: User enters their CNIC and Password.
- Step 3: User clicks the "Login" button.
- Step 4: The system validates the credentials against the Users table in the database.
- Step 5: If credentials are valid, the system generates a JWT token and redirects the user to their role-specific dashboard based on their Role field (Admin, Member, President, Vice President, Treasurer).
- Step 6: If credentials are invalid, the system displays an error message: "Invalid CNIC or Password".

Annexure B: User Dashboards

Upon successful login, users are redirected to their role-specific dashboard. Below is a description of each dashboard.

Annexure C: Member Dashboard

The Member Dashboard provides access to the following features:

- Submit Complaint: Submit a new complaint with title, description, and optional images.
- My Complaints: View the status of all submitted complaints (Pending, In Progress, Committee Resolved, Approved, Rejected, Resolved).

- Elections: View active elections and vote for candidates (President, Vice President, Treasurer).
- My Committees: View committees the member is assigned to.
- Change Council: Request to change council membership.
- Notifications: View all notifications received.
- Suggestions: Submit suggestions to the council.

Annexure D: Admin Dashboard

The Admin Dashboard provides access to the following features:

- Create Council: Create a new neighborhood council using Google Maps integration.
- Manage Councils: View and manage all existing councils.
- Announce Election: Set election start date, end date, and nomination period.
- Change Requests: View and manage council change requests submitted by members.

Annexure E: President Dashboard

The President Dashboard provides access to the following features:

- Complaints: View all complaints within the council, filtered by status.
- Form Committee: Select a complaint and form a committee to resolve it.
- My Committees: View all committees formed and their status.
- Review Reports: View committee reports including meeting minutes and resolution images.
- Approve/Reject Complaints: Review committee reports and approve or reject complaints.
- Approve Budget: Approve budget requests for complaint resolution.

Annexure F: Database Schema

The database schema for the Neighborhood Council App consists of 23 tables designed using Microsoft SQL Server. Key entities include:

- **USERS:** Stores user credentials, CNIC, role, and profile details.
- **NHC_ZONES:** Stores neighborhood council boundaries and details.
- **USERNHCS:** Manages multi-council membership for users.
- **COMPLAINTS:** Stores complaint details, status, and associated committee/president remarks.
- **COMPLAINT_IMAGES:** Stores image paths for complaints.
- **COMMITTEES:** Stores committee details formed for complaint resolution.
- **COMMITTEE_MEMBERS:** Maps members to specific committees.
- **COMPLAINT_MEETINGS:** Stores PDF paths for meeting minutes and remarks.
- **COMPLAINT_RESOLUTION_IMAGES:** Stores images proving complaint resolution.
- **ELECTIONS:** Manages election schedules and statuses.
- **CANDIDATES:** Stores candidate profiles for elections.
- **NOMINATIONS:** Manages member nominations for positions.
- **PANELS:** Stores registered election panels.
- **ELECTION_VOTES:** Securely records votes cast by members.
- **COUNCIL_TREASURY_FUNDS:** Manages total, released, and available funds per council.
- **COUNCIL_TREASURY_FUND_RELEASES:** Maintains a transparent ledger of all fund releases.
- **COMPLAINT_BUDGET_REQUESTS:** Manages budget requests for

approved complaints.

- REQUESTS: Stores member requests for council changes.
- NOTIFICATIONS: Logs real-time notifications sent to users.
- SUGGESTIONS: Stores suggestions submitted by members.